

Hello again folks!

Thank you for dropping in for Python 101: Starting from Scratch. I hope you enjoyed the workshop. Please fill out the survey sent out by iSkills as it helps me keep the workshops running well. You can find slides for Python 101 [here](#). Before we head into Python 102: Building on the Basics, it'd be a good idea to cover functions since we weren't able to practice in class.

Functions

A quick refresher, functions are blocks of code that we write for programming tasks that we do repeatedly. In order to use a function of our making, we must first *define* a function. This is simply writing a function for the first time. This is how it looks:

A function takes in **inputs** and then returns an **output**

It follows the following pattern,

```
def function (parameter, parameter):
```

```
    [do something]
```

```
    return result
```



Indentation is important! It indicates where an code block ends. Hit the tab button to do this

Here def and return are **keywords**!

We use the def keyword for defining a function and we place it before the name of the function and its parameters in brackets. The return keyword gives the result to the place where we call the function. What's calling a function? Well, once the function has been defined, we simply *call* it which is the technical way of saying, we use the function in some place in our code.

Let's say we want to have a function that checks if we have enough tomatoes in our pantry. Let's first define such a function:

```
def tomato_checker(count_of_tomatoes):  
    if count_of_tomatoes > 5:  
        return("We have enough tomatoes")  
    else:  
        return("We don't have enough tomatoes")
```

Now that we've defined the function, we can call it. Let's say I want to store the result of the function into a string variable and then print it. This is what that would look like.

```
tomatoes = 4  
message = tomato_checker(tomatoes) #this is calling the function  
print(message)
```

We don't have enough tomatoes

Homework

My mom's butter chicken recipe calls for 5 onions. I have 3 red onions and 1 white onion. Is that enough?

I need you to write me a function that has two parameters. These parameters will be the number of red onions and number of white onions in our pantry. I want the function to tell me if I have enough total onions, regardless of type, for my recipe. Define the function, create two variables for the two onion types. Lastly, pass these variables into the function call and print a message whether or not I have enough.

Send your code and result to omer.imran@mail.utoronto.ca and I'll check if it's correct. Good luck!

